Part I: Decisions, distributions, data

Predictions: Theory and Practice

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Optimal Predictions

Example: Good Judgement

- Good judgement (link)
  - webpage gathering (and selling) predictions
  - awarding the title of “superforecasters” (top 2%)
- The Economist 2023: “The Good Judgment team [...] correctly forecasting the outcomes of eight questions...”
  - Global growth was 3%
  - China grew by 5%
  - ruling-party candidates won in Nigeria and Turkey
  - Vladimir Putin was not ousted
  - there was no election in Britain
  - no clash over Taiwan
  - no nuclear device detonated by Russia

Current Topics on Good Judgement Open (2026)

- **Global Armed Conflict**
  - Will Mojtaba Khamenei flee Iran or cease to be its supreme leader before Sep 2026?
- **Artificial Intelligence**
  - What will be the AI model adoption rate for large US businesses in Dec 2026?
  - What will be NVIDIA's total revenue in Q1 of fiscal year 2027?
- **The Economist: World Ahead 2026** (20 questions)
  - Will the FDA approve Moderna's mRNA flu vaccine before Oct 2026?
  - India's real GDP growth (2026-27)
  - Effective US tariff rate for all imports in H2 2026
- **"Right!" said FRED**
  - Spread between 10-year and 2-year Treasury yields
  - 30-year fixed mortgage rate for week ending 25 June 2026
  - unemployment rate for June 2026

Goals

By the end of this lecture, you should be able to answer:

1. What are the main **types** of prediction targets?
2. When is a prediction **accurate**?
3. When is a prediction **calibrated**?
4. When is a prediction **sharp**?
5. In which sense is calibration **optimal**?

Mathematics motiviation for everyone

- this lecture is very mathematical
- my take on mathematics:
  - mostly about developing **intuition that is government by logic**
  - two step procedure:
    1. intuition: “A seems true.”
    2. mathematics: “Check A, if true proceed, otherwise revise intuition.”
  - developing this intuition requires exposure
  - LLMs can handle math for you, but often struggle with transfer to applications
- we will revisit the concepts when empirically analyzing predictions and human behavior

Types of predictions

Targets of predictions

- the three most important targets of predictions are
  - binary events
    - $\{0, 1\}$
    - TRUE,FALSE
    - event E, complement E^C
    - examples: $\{rain, norain\}$, {higher, not higher}
  - categorical variables
    - $\{E_1, E_2, \ldots, E_k\}$
    - examples: countries, $\{rock, paper, scissors\}$, {higher, same, lower},...
  - real-valued (continuous) variables
    - $y \in \mathbb{R}$
    - can be decomposed into infinite number of events ($y \leq c$)
    - examples: money values, length, weight, ...

- we start with **binary events**
  - categorical and real-valued can be reduced to event predictions
  - later concepts will be generalized to real-valued prediction targets

Necessary knowledge

- a random variable y
  - maps into the space of **possible outcomes**
  - example (coin toss): maps into {“heads”,“tails”}
  - takes on one **realized value** (also “realization”)
- distribution of a random variable
  - distribution over the possible outcomes
  - example (coin toss): P(“head”) = 50%, P(“tails”) = 50%
- Sidenote: not necessary are the concepts of
  - measureability of a random variable
  - σ-algebras as a concept for information sets

- for a real-valued random variable y...
  - possible outcomes are numbers
  - example (coin toss): with "heads" = 1, "tails" = 0
- ... the **expectation** or **expected value** $E[y]$
  - is a real number.
  - is a property of the distribution, not of the realized value
- example (coin toss):
  - possible outcomes: "heads" = 1, "tails" = 0
  - distribution / probabilities: $P(\text{" 0 "}) = P(\text{" 1 "}) = 50\% = 0.5$
  - Expected value

$$E[y] = 0 \times P(\text{" 0 "}) + 1 \times P(\text{" 1 "}) = 0 \times 0.5 + 1 \times 0.5 = 0.5$$

Point (deterministic) predictions

Definition: point predictions.

A prediction $\hat{y}$ for an event is a point prediction if it is an element of the potential outcomes $\{0, 1\}$.

- no direct connection to decision-making
  - remember: Decisions are informed by probabilities

Probabilistic predictions

Definition: probabilistic predictions.

The prediction $\hat{p}$ is called probabilistic if it is a **probability distribution** over the elements $\{0, 1\}$.

- distribution over $\{0, 1\}$ identified by $p = P(1)$
  - follows that $P(0) = 1 - p$

What is a good prediction (in hindsight)?

- easy, if y = 1 then
  - $\hat{y} = 1$ is better than $\hat{y} = 0$
  - $\hat{p} = 1$ is better than any $\hat{p} < 1$
- but this is **ex-post** (in hindsight)
- How does a **good prediction** look like in general?
  - before the uncertainty around the target y is resolved
  - we need a more technical framework to handle this question!

The prediction space setting

- technically we understand the prediction target y and the prediction $\hat{y}$ or $\hat{p}$ as **random variables**
  - this is called the **prediction space setting** (Gneiting and Katzfuss 2014)
  - we judge the performance of a prediction via the relation between the random variables
- what would be good properties for a prediction?
  - perfect $\hat{y} = y$ (often unrealistic)
  - unbiased $E[\hat{y} - y] = 0$ (classic property)
  - error $e = y - \hat{p}$ should be small (note: e is also a random variable)
    - e.g., mean absolute error $E[|e|]$ small

Notation summary

- uncertain outcome y
- its (unknown) distribution p
  - we write $E_{y \sim p}[f(y)]$ or shorter $E_p[f(y)]$
  - $E[f(y)]$ normally means $E_{y \sim p}[f(y)]$
- a (point) prediction $\hat{y}$
- a probability prediction $\hat{p}$
- Example:
  - y is income increasing next year.
  - $\hat{y}$ could be "Yes".
  - $\hat{p} = 0.8 = 80\%$.

Toy example

- Let's consider a toy example: Target $y \sim Bernoulli(0.5)$

R | Python

```r
# a random sample from the Bernoulli distribution
set.seed(1)
rbinom(n = 10, p = 0.5, size = 1)
```

```
[1] 0 0 1 1 0 1 1 1 1 0
```

We will use these five cases as running examples throughout this section and compare them head-to-head at the end:

- perfect prediction $\hat{p} = y$
- unbiased prediction $\hat{p} = 0.5$
- biased prediction $\hat{p} = 0.6$
- unbiased prediction with noise $\hat{p} \sim U[0, 1]$ (and independent of y)
- unbiased prediction with maximum noise $\hat{p} = 1 - y$ (depends on y)

The prediction space setting illustrated

- Next we discuss two scenarios for which the prediction space setting is appropriate:
  1. **sampling predictions from data** — $\hat{p}$ varies because data varies
  2. **sampling from a population of prediction problems** — $\hat{p}$ varies across different forecasters or events
  - in which sense is prediction $\hat{p}$ a random variable?

… sampling predictions from data

- one approach to predict an event y:
  - take a sample of data $(y_1, \ldots, y_n)$
  - compute **sample average** $\hat{p} = \frac{1}{n} \sum_{i=1}^{n} y_i$
- if(!) data is **randomly sampled** from p, the prediction $\hat{p}$
  - is unbiased
  - increases in accuracy with sample size n
- "sampling uncertainty": $\hat{p}$ is a random variable as a function of random data

… sampling from prediction problems

- there is another way to interpret the prediction space setting
- consider a population of prediction problems, where each draw consists of the pair (target outcome, prediction)

$$(y, \hat{y}) \text{ or } (y, \hat{p})$$

- Example: All the predictions generated from the CIA
  - our sample consists of all the recorded predictions
  - one observation might look like
    - Event: Does Russia annex Crimea in 2014?
    - prediction: “No” or “5%”
  - In the prediction space setting we can answer questions like
    - which expert in the CIA is doing best
    - what systemic mistakes can be detected from the data
    - how can we improve a raw prediction

Example: Yourself

- try to write down (subconscious) beliefs as predictions
- example targets
  - I will finish this project until xx.xx.xxxx. (too optimistic?)
  - Will I enjoy working on an exercise with a random student from this class? (too pessimistic?)
  - Next week, I will start to do xxx. ("planning you" more disciplined than "implementing you"?)

Conditional expectation: a reminder

- ...**conditional expectation** given $x = c$,

$$E[y|x = c]$$

  - expected value with respect to conditional distribution $P(y|x = c)$
  - known with Bayes rule
  - is a real number
- ...**conditional expectation**,

$$E[y|x]$$

  - is a function of x
  - is a random variable

- Rules for the conditional expectation
  - linear

$$E[y_1 + y_2|x] = E[y_1|x] + E[y_2|x]$$

$$E[cy|x] = cE[y|x] \text{ for } c \text{ constant}$$

  - tower property

$$E[E[y|x]] = E[y]$$

    - *averaging a conditional expectation over all conditions gives back the unconditional expectation — double averaging collapses to single averaging*
  - or more general

$$E[E[y|x_1, x_2]|x_1] = E[y|x_1]$$

  - Functions f of conditioning information are constant:

$$E[f(x)|x] = f(x)$$

$$E[f(x)y|x] = f(x)E[y|x]$$

  - if y and x independent, then

Calibration

Definition: Calibration

A prediction $\hat{p}$ is **calibrated** for y if

$$E(y|\hat{p}) = \hat{p}.$$

- The calibration condition is equivalent to $P(y = 1|\hat{p}) = \hat{p}$
- Calibration means that after observing a prediction (but nothing else), we cannot improve the prediction
- can be expressed as property of the forecast error $e = y - \hat{p}$:

$$E(y|\hat{p}) = \hat{p} \Leftrightarrow E(y - \hat{p}|\hat{p}) = 0$$

Thus, calibration means that the forecast error is zero for all predicted values $\hat{p}$.

Calibration vs. Unbiasedness

- **Unbiasedness**: a *global* (marginal) condition on the average prediction

$$E[y] = E[\hat{p}]$$

- **Calibration**: a *local* (conditional) condition for every value of $\hat{p}$:

$$E[y|\hat{p}] = \hat{p}$$

Calibration implies unbiasedness, but not vice versa.

- Calibration $\Rightarrow$ unbiasedness: by the tower property

$$E[y] = E[E[y|\hat{p}]] = E[\hat{p}]$$

- Unbiasedness $\not\Rightarrow$ calibration — counterexample:
  - $y \sim Bernoulli(0.5), \hat{p} \sim U[0,1]$ independent of y
  - Unbiased: $E[\hat{p}] = 0.5 = E[y]$ ✓
  - Not calibrated: $E[y|\hat{p}] = 0.5 \neq \hat{p}$ for most values of $\hat{p}$ ✗

- One example of a calibrated prediction is the random sample predictor
  - assume we can sample from similar events (e.g., “does webpage visitor buy item A”)
  - predict next visitor with **sample average**

$$\hat{p} = \frac{1}{n} \sum_{i=1}^{n} y_i$$

  - this prediction is calibrated

Proof (handwritten in lecture):

random sample predictor

If $y_i \sim_{iid} p$ then $\hat{p} = \frac{1}{n}\sum_{i=1}^{n} y_i$ is a calibrated prediction for y_{n+1}.

- Proof:

$$
\begin{aligned}
E[y - \hat{p}\,|\hat{p}, p] =& E[y - \frac{1}{n}\sum_{i=1}^{n} y_i|\hat{p}, p] \\
=& E[y|\hat{p}, p] - E[\frac{1}{n}\sum_{i=1}^{n} y_i|\hat{p}, p] = p - \frac{1}{n}np = 0
\end{aligned}
$$

and thus

$$
E[y - \hat{p}\,|\hat{p}] = E[\underbrace{E[y - \hat{p}\,|\hat{p}, p]}_{=0}|\hat{p}] = 0
$$

Calibration curve

- We can understand calibration as a property of the **calibration curve** defined by the conditional expectation function

$$f(\hat{p}) = E[y|\hat{p}].$$

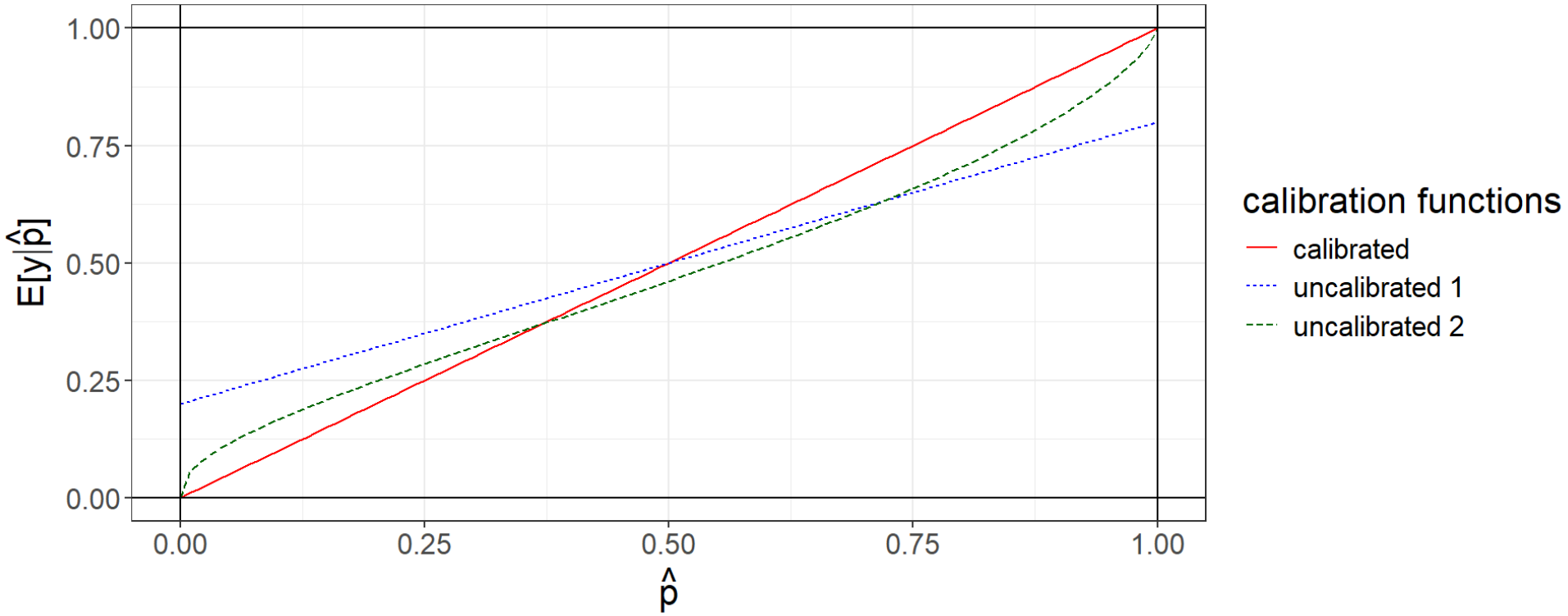


Figure 1: Some calibration curves. The red line is calibrated with $f(\hat{p}) = \hat{p}$.

Summary: Calibration

- A prediction is **calibrated** if

$$E(y|\hat{p}) = \hat{p}.$$

- The ratio of events happening in a **random sample** is a calibrated prediction
- The **calibration curve** explains deviations from calibration
- Open question: How does calibration relate to **decision-making**?

Accuracy

- consider the situation with two predictions (e.g., two experts) for the same event
  - who is the better prediction expert?
- if you know your decision problem, the best expert maximizes your utility
  - based on a prediciton $\hat{p}$ you take the action that maximizes your expected utility

$$a^\star(\hat{p}) = argmax_a E_{y \sim \hat{p}}[u(a, y)]$$

  - so the better expert leads to higher expected utilies

$$U(\hat{p}) = E[u(a^\star(\hat{p}), y)]$$

  - the highest expected utility can be achieved if $\hat{p} = p$ where p is the actual distribution of y

Proper scoring rules

- there are also generic functions $s(\hat{p}, y)$ with the property that their **expected utility** is **maximized at the true distribution** of y

Definition: Proper scoring rule.

A function $s(p, y)$ is called a **scoring rule**.

A scoring rule is called **proper** if $f(\hat{p}) = E[s(\hat{p}, y)]$ has a maximum at $\hat{p} = p$.

- sidenotes:
  - Sometimes difference between proper and *strictly* proper
  - Sometimes oriented differently such that proper means minimizing expected score ("loss function")

Quadratic scoring rule

- the most famous proper scoring rule is the **quadratic scoring rule** (also called **Brier score**)

Definition: The quadratic scoring rule.

$$s_{qsr}(\hat{p}, y) = 1 - (\hat{p} - y)^2$$

- The quadrtic scoring rule

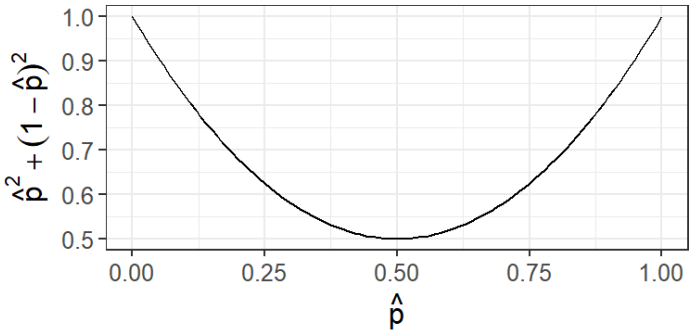
$$s_{qsr}(\hat{p}, y) = 1 - (\hat{p} - y)^2$$

is equivalent to

$$s_{qsr}(\hat{p}, y) = 2\hat{p}_y - \hat{p}^2 - (1 - \hat{p})^2$$

where $\hat{p}_y = \hat{p}$ if $y = 1$ and otherwise $\hat{p}_y = (1 - \hat{p})$

  - the first term $2\hat{p}_y$ rewards probability put on the true event
  - the second term penalizes putting mass on the extremes



Proof (handwritten in lecture):

The QSR is proper

Proof: The QRS is proper.

Look at the FOC of $E[s_{qsr}(\hat{p}, y)]$ to find optimal $\hat{p}$

$$\partial\hat{p}\, E[s_{qsr}(\hat{p}, y)] = E[\partial\hat{p}\, s_{qsr}(\hat{p}, y)] = E[-2(\hat{p} - y)] = -2(\hat{p} - E[y])$$

and

$$\partial\hat{p}\, E[s_{qsr}(\hat{p}, y)] = 0 \iff \hat{p} = E[y]$$

Thus, the expectation is maximized if true propability of y is issued as prediction, i.e., the quadratic scoring rule is proper.

(Using: $E[y] = p \times 1 + (1 - p) \times 0 = p$.)

What is wrong with a linear score?

- the most intuitive idea would be to consider scores

$$s_l(\hat{p}, y) = |y - \hat{p}|$$

The expected score is:

$$\mathbb{E}[s(\hat{p}, y)] = p \cdot |1 - \hat{p}| + (1 - p) \cdot |\hat{p} - 0| = p(1 - \hat{p}) + (1 - p)\hat{p}$$

Simplifying:

$$\mathbb{E}[s(\hat{p}, y)] = p - p\hat{p} + \hat{p} - p\hat{p} = p - 2p\hat{p} + \hat{p}$$

Taking the derivative with respect to $\hat{p}$:

$$\frac{d}{d\hat{p}}\mathbb{E}[s(\hat{p}, y)] = -2p + 1$$

This derivative is constant in $\hat{p}$, indicating that the expected score is linear in $\hat{p}$:

(If $p = \frac{1}{2}$, the expected score is constant, so any $\hat{p} \in [0, 1]$ minimizes the score.)

- Thus, the linear score is not proper. It incentivizes to report the event with the higher probability (a point prediction).

The logarithmic score

- another widely used proper scoring rule is the **logarithmic scoring rule** (also called **log score**)

Definition: The logarithmic scoring rule.

$$s_{log}(\hat{p}, y) = log(\hat{p}_y),$$

where $\hat{p}_y$ is probability mass of $\hat{p}$ at the realized target value y, i.e. $\hat{p}_1 = \hat{p}$ and $\hat{p}_0 = 1 - \hat{p}$.

- It is no coincidence that you know the quadratic score/loss from OLS and the logarithmic score from Maximum Likelihood estimation.

Summary: Accuracy

- How do we compare two predictions (experts or systems)?
  - If we have a decision and a utility we use the expected utility

$$U(\hat{p}) = E[u(a^\star(\hat{p}), y)]$$

  - If we do not have the specifics, we can use proper scoring rules

$$S(\hat{p}) = E[s(\hat{p}, y)]$$

- We can call a prediction $\hat{p}_A$ **more accurate than** $\hat{p}_B$ if

$$S(\hat{p}_A) > S(\hat{p}_B)$$

- One proper scoring rule is the quadratic scoring rule

$$s_{qsr}(\hat{p}, y) = 1 - (y - \hat{p})^2$$

Sharpness

- sharpness describes the concentration of a probabilistic prediction
  - not applicable to point predictions (or point predictions are the sharpest of predictions)
- sharpness is a property of the prediction alone
  - not of prediction and target
- Sometimes also referred to as “boldness”

The variance of a distribution

- **The variance of the predictive distribution** is a natural **measure of sharpness**
- the variance of a distribution is defined as

$$\mathrm{var}_{y\sim p}(y) = E_p[(y - E_p[y])^2]$$

- For a probability p for an event this is

$$\mathrm{var}_{y\sim p}(y) = p(1-p)^2 + (1-p)p^2 = p(1-p)[(1-p)+p] = p(1-p)$$

- For a prediction $\hat{p}$, we rescale and define

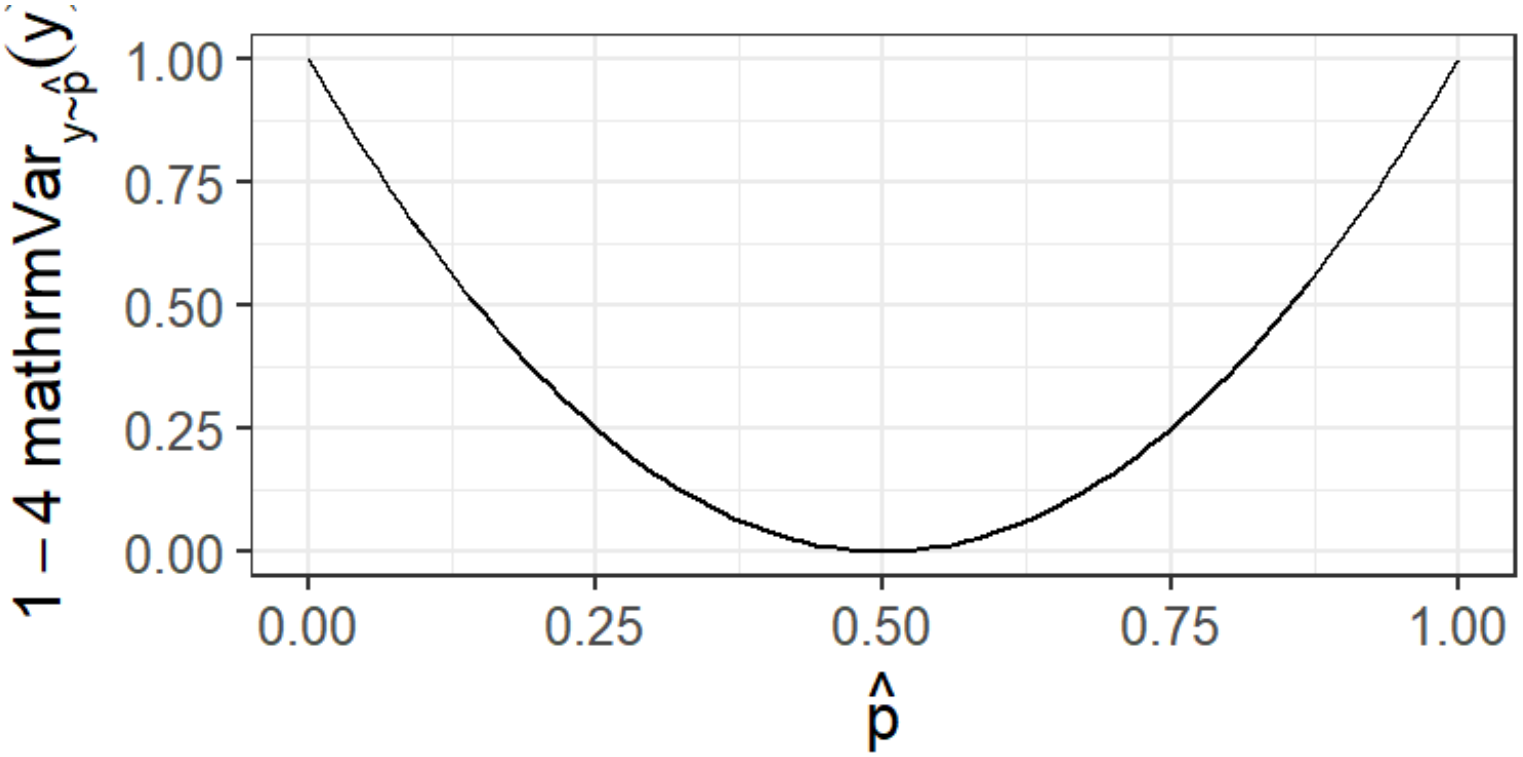
$$\mathrm{sharpness}(\hat{p}) = 1 - 4\,\mathrm{var}_{y\sim\hat{p}}(y)$$

> $\mathrm{var}_{y\sim\hat{p}}(y) = \hat{p}(1-\hat{p})$ is the variance of **outcome** y drawn from the predictive distribution $\hat{p}$ — not the variance of $\hat{p}$ as a random variable across predictions.

Illustrating sharpness

$$\text{sharpness}(\hat{p}) = 1 - 4\,\text{var}_{y\sim\hat{p}}(y)$$

- High sharpness means the predictions are close to 0 or 1, indicating strong confidence in the outcome.
- Low sharpness means predictions are closer to 0.5, showing less confidence and more uncertainty.



Expected score and variance

- Sharpness can be interpreted in terms of scores and utilities
- Consider the expected score under probability p

$$S(p) = E_p[s(p, y)]$$

  where E_p means the expectation assuming $y \sim p$.
- If score s is derived from decision and utility, then

$$S(p) = \text{ expected utility under uncertainty } p$$

- for the QSR:

$$\begin{aligned} S_{qsr}(p) &= E_p[s_{sqr}(p, y)] \\ &= 1 - p(1-p)^2 + (1-p)p^2 \\ &= 1 - p(1-p)[(1-p) + p] = \\ &= 1 - var(p) \end{aligned}$$

Toy example revisited

prediction	Miscalibration function $\lvert E[y\lvert\hat{p}] - \hat{p}\rvert$	expected score $E[s_{qsr}(\hat{p}, y)]$	Sharpness $1 - 4Var(\hat{p})$
$\hat{p} = y$	0	1	1
$\hat{p} = 0.5$	0	0.75	0
$\hat{p} = 0.6$	0.1	0.74	0.04
$\hat{p} \sim Bern(0.5)$	0.5	0.5	1

Note: $\hat{p} \sim Bern(0.5)$ achieves maximum sharpness (1) because it only takes the extreme values 0 and 1 — yet it is badly miscalibrated because it is independent of y. Sharpness alone is not enough.

Sidenote: Entropy

- if we take the negative expected score with respect to the **logarithmic scoring** rule $s_{log}(p, y) = log(p_y)$ we get the **entropy**
- an information measure describing **randomness** or **uncertainty**
  - the average amount of information required to describe the outcome of a random event
  - measured in bits
- also a physical concept describing the **disorder** of a system
  - governed by the "second law of thermodynamics"
  - total entropy of an isolated system tends to increase over time

Optimal: Sharpness subject to calibration

In summary:

- Sharpness = Concentration of predictions (confidence level)
- Calibration = Agreement between predictions and actual outcomes

Good probabilistic forecasts aim to be both sharp and well-calibrated.

Decomposition of accuracy

- The performance of a prediction in terms of a scoring rule can be decomposed into three parts
  - define $p_C = E[y|\hat{p}]$ a calibrated version of $\hat{p}$
  - define $p_R = E[y]$ a constant calibrated prediction (contains no information)
  - define $S(\hat{p}) = E[s(\hat{p}, y)]$
- Then the expected score is

$$\begin{aligned} S(\hat{p}) &= S(p_R) + [S(p_C) - S(p_R)] - [S(p_C) - S(\hat{p})] \\ &= \underbrace{S(p_R)}_{\text{UNC}} + \underbrace{[S(p_C) - S(p_R)]}_{\text{DSC}} - \underbrace{[S(p_C) - S(\hat{p})]}_{\text{MSC}}, \end{aligned}$$

- UNC (uncertainty): score from uncertainty in outcome
- DSC (discrimination ability): score improvement by prediction
- MSC (miscalibration): score loss because prediction is miscalibrated

- Then the expected score is

$$S(\hat{p}) = \mathrm{UNC} + \mathrm{DSC} - \mathrm{MSC},$$

- We want MSC to be small (ideally $\mathrm{MSC} = 0$) and DSC large
- When is DSC large?
  - if $\hat{p}$ far away from $p_r = E[y]$, i.e. if $\hat{p}$ is bold/sharp.
- In this sense, we want to maximize sharpness subject to calibration.

ideal predictions and information

- Another way to look at this is via information/input variables
- let's say x denotes all the information necessary to predict y perfectly
  - some part of x is known to the forecaster, x_k
  - the remainder, say x_u, is unknown to the forecaster
  - we have $x = (x_k, x_u)$
- the prediction is a function of the known information
  - $\hat{p} = f(x_k)$

The conditional expectation is optimal.

Among all functions of x_k, the conditional expectation

$$\hat{p} = E[y|x_k]$$

maximizes any expected proper score $E[s(\hat{p}, y)]$ (Holzmann and Eulert 2014)

- $E[y|x_k]$ is also called the **ideal prediction** for information x_k

Ideal implies calibrated.

All ideal predictions are calibrated.

- Proof:

$$E[y - E[y|x_k]|\hat{p}] = E[y|\hat{p}] - E[E[y|x_k]|\hat{p}] = 0$$

(as $E[E[y|x_k]|\hat{p}] = E[y|\hat{p}]$ because $\hat{p} = f(x_k)$)

Summary: Optimal Predictions

- the prediction space setting understands target y and prediction $\hat{p}$ as random variables with a joint distribution
- Properties of a prediction are
  - sharpness: $1 - 4Var(\hat{p})$
  - calibration: $E[y|\hat{p}] = \hat{p}$
  - idealness: $\hat{p} = E[y|x]$ for some information x
  - accuracy: $E[s(\hat{p}, y)]$ for some score s or utility function such that $s = u(a^\star(\hat{p}), y)$
- A good prediction maximizes sharpness and calibration.
- Open question: What to do if there is a trade-off between sharpness and calibration?
  - Machine learning predictions are often uncalibrated!

References

Gneiting, Tilmann, and Matthias Katzfuss. 2014. “Probabilistic Forecasting.” *Annual Review of Statistics and Its Application* 1 (1): 125–51.
Holzmann, Hajo, and Matthias Eulert. 2014. *The Role of the Information Set for Forecasting—with Applications to Risk Management*.